

DATE:

EXPT NO: 02

EDA - Data Import and Export

AIM:

To perform Exploratory Data Analysis (EDA) and data cleaning on the Earthquake dataset to identify patterns, handle missing values, remove duplicates, and prepare the data for Machine Learning.

PROCEDURE:

1. Import the required Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.
2. Load the Earthquake dataset using Pandas.
3. Display the first few records and check the dataset shape.
4. Examine the data types and dataset information.
5. Check for missing or null values.
6. Handle missing values using suitable methods.
7. Identify and remove duplicate records.
8. Check for incorrect or inconsistent data.
9. Perform statistical analysis using describe().
10. Visualize important features using graphs and charts.
11. Analyze correlations between numerical features using a heatmap.
12. Save the cleaned dataset for further analysis and Machine Learning.

CODE:

```
import pandas as pd

# Import Excel file
df = pd.read_excel(r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx")

# Display first 5 records
print("First 5 Records")
print(df.head())

# Dataset Information
print("\nDataset Information")
df.info()

# Statistical Summary
print("\nStatistical Summary")
```

```
print(df.describe(include='all'))
```

OUTPUT:

First 5 Records

										title	magnitude	date_time	cdi	mml	alert
tsunami	sig	net	nst	dmingap		magType	depth	latitude	longitude						
location		continent		country											
0	M 7.0 - 18 km SW of Malango, Solomon Islands						7.0	22-11-2022 02:03	8	7	green				
1	768 us 117 0.509 17.0				mwW	14.000	-9.7963	159.596	Malango, Solomon Islands						
Oceania Solomon Islands															
1	M 6.9 - 204 km SW of Bengkulu, Indonesia						6.9	18-11-2022 13:37	4	4	green				
0	735 us 99 2.229 34.0				mwW	25.000	-4.9559	100.738	Bengkulu, Indonesia						
NaN		NaN													
2	M 7.0 -						7.0	12-11-2022 07:09	3	3	green				
1	755 us 147 3.125 18.0				mwW	579.000	-20.0508	-178.346			NaN				
Oceania		Fiji													
3	M 7.3 - 205 km ESE of Neiafu, Tonga						7.3	11-11-2022 10:48	5	5	green				
1	833 us 149 1.865 21.0				mwW	37.000	-19.2918	-172.129	Neiafu, Tonga						
NaN		NaN													
4	M 6.6 -						6.6	09-11-2022 10:14	0	2	green				
1	670 us 131 4.998 27.0				mwW	624.464	-25.5948	178.278			NaN				
NaN		NaN													

Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 782 entries, 0 to 781

Data columns (total 19 columns):

#	Column	Non-Null	Count	Dtype
0	title	782	non-null	object
1	magnitude	782	non-null	float64
2	date_time	782	non-null	object
3	cdi	782	non-null	int64
4	mmi	782	non-null	int64
5	alert	415	non-null	object
6	tsunami	782	non-null	int64
7	sig	782	non-null	int64
8	net	782	non-null	object
9	nst	782	non-null	int64
10	dmin	782	non-null	float64
11	gap	782	non-null	float64
12	magType	782	non-null	object
13	depth	782	non-null	float64
14	latitude	782	non-null	float64
15	longitude	782	non-null	float64
16	location	777	non-null	object
17	continent	206	non-null	object
18	country	484	non-null	object

```
dtypes: float64(6), int64(5), object(8)
```

```
memory usage: 116.2+ KB
```

Statistical Summary

	magnitude	cdi	mmi	tsunami	sig	nst	dmin
gap	depth	latitude	longitude				

```

count 782.000000 782.000000 782.000000 782.000000 782.000000 782.000000 782.000000
782.000000 782.000000 782.000000 782.000000
mean 6.940153 4.333760 5.964194 0.388747 776.008951 275.396419 2.231937
25.807864 75.883168 2.386000 39.734707
std 0.445213 2.658939 1.983592 0.487778 185.083392 218.192074 2.839655
17.652131 137.279603 30.344421 108.016952
min 6.500000 0.000000 1.000000 0.000000 650.000000 0.000000 0.005000
9.000000 9.000000 -61.848000 -179.962000
50% 6.800000 4.000000 6.000000 0.000000 708.500000 213.000000 1.300000
21.000000 26.000000 -3.580000 103.982000
75% 7.100000 7.000000 7.000000 1.000000 819.750000 385.000000 2.880000
29.000000 55.000000 26.126500 139.710250
max 9.100000 9.000000 9.000000 1.000000 2910.000000 934.000000 17.654000
239.000000 670.810000 71.631200 179.662000
[11 rows x 19 columns]

```

```

import pandas as pd

df = pd.read_csv(r"C:\Users\babyn\Downloads\archive\earthquake_data.csv")
print(df.head())

```

OUTPUT:

```

              title magnitude      date_time cdi mmi alert
tsunami sig net nst  dmin gap magType  depth latitude longitude
location continent          country
0  M 7.0 - 18 km SW of Malango, Solomon Islands      7.0 22-11-2022 02:03    8    7 green
1  768 us 117 0.509 17.0      mww 14.000 -9.7963 159.596 Malango, Solomon Islands
Oceania Solomon Islands
1      M 6.9 - 204 km SW of Bengkulu, Indonesia      6.9 18-11-2022 13:37    4    4 green
0  735 us  99 2.229 34.0      mww 25.000 -4.9559 100.738  Bengkulu, Indonesia
NaN                      NaN
2
1  755 us 147 3.125 18.0      mww 579.000 -20.0508 -178.346
Oceania                      Fiji
3      M 7.3 - 205 km ESE of Neiafu, Tonga      7.3 11-11-2022 10:48    5    5 green
1  833 us 149 1.865 21.0      mww 37.000 -19.2918 -172.129  Neiafu, Tonga
NaN                      NaN
4
1  670 us 131 4.998 27.0      mww 624.464 -25.5948 178.278
NaN                      NaN

[5 rows x 19 columns]

```

```

import pandas as pd
import sqlite3

# Read Excel file
df = pd.read_excel(r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx")

```

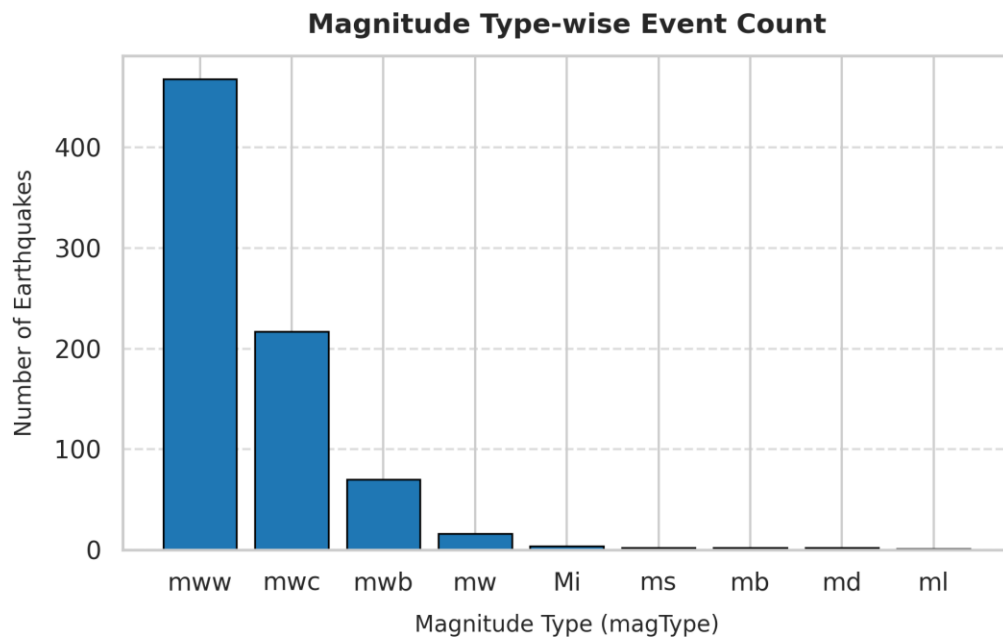
```
# Connect to SQLite database
conn = sqlite3.connect("earthquake_data.db")

# Export data to SQLite table
df.to_sql("earthquake_data", conn, if_exists="replace", index=False)
print("Data exported to SQLite successfully!")
conn.close()
```

```
import matplotlib.pyplot as plt

mag_type = df["magType"].value_counts()
plt.bar(mag_type.index, mag_type.values)
plt.title("Magnitude Type-wise Event Count")
plt.xlabel("Magnitude Type (magType)")
plt.ylabel("Number of Earthquakes")
plt.show()
```

OUTPUT:



RESULT: